

Business Statistics - MS6107E

Dr. Arjun Anil Kumar

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Outline – Part I

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 - Non-Probabilistic Sampling
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Sampling

Variable

Variable

A variable is a measurable characteristic that can assume different values among subjects or instances (e.g., age, height, income, gender).

Variables are of different types -

- Nominal
- Ordinal
- Ratio
- Interval



Types of Variables

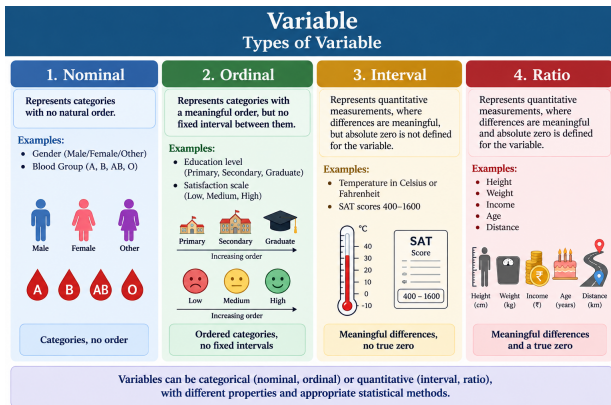


Figure: Types of Variables



Variable

Types of Variable

- **Nominal** variables represent categories with no natural order.
For eg : Gender (Male/Female/Other), Blood Group (A, B, AB, O).
- **Ordinal** variables represent categories with a meaningful order, but no fixed interval between them.
For eg : Education level (Primary, Secondary, Graduate),
Satisfaction scale (Low, Medium, High).
- **Interval** variables represent quantitative measurements, where differences are meaningful, but absolute zero is not defined for the variable.
For eg : Temperature in Celsius or Fahrenheit, SAT scores 400-1600
- **Ratio** variables represent quantitative measurements, where differences are meaningful and absolute zero is defined for the variable.
For eg : Height, Weight, Income, Age and Distance.



Sampling Methods

Sampling

The process of selecting a subset (sample) from a population to draw conclusions about the entire population. The goal of sampling is to obtain representative data in a cost-effective and time-saving way.

Sampling is of different types

- **Probabilistic or Random Sampling**
- **Non-probabilistic Sampling**



Probabilistic Sampling

Every element of the population has a known, non-zero probability of being included in the sample.

- **Simple Random Sampling** : Every unit has an equal chance.
For eg : Lottery draw, random number generator.
- **Systematic Sampling** : Select every k-th element after a random start.
For eg : Survey every 10th customer entering a mall.
- **Stratified Sampling** : Divide population into strata (groups), then sample randomly within each.
For eg : Divide students by year (UG, PG, PhD) and sample proportionally.
- **Cluster Sampling** : Divide population into clusters (often geographically), then randomly choose some clusters and study all units inside them.
For eg : Select 5 villages randomly and survey all households within them.



Random Sampling with and without Replacement

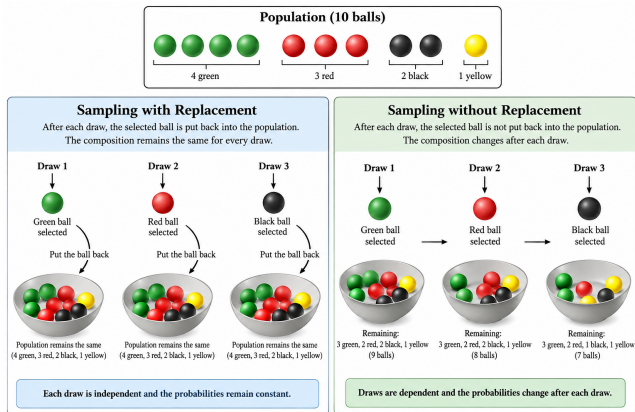


Figure: Random Sampling with and without Replacement



Systematic Sampling

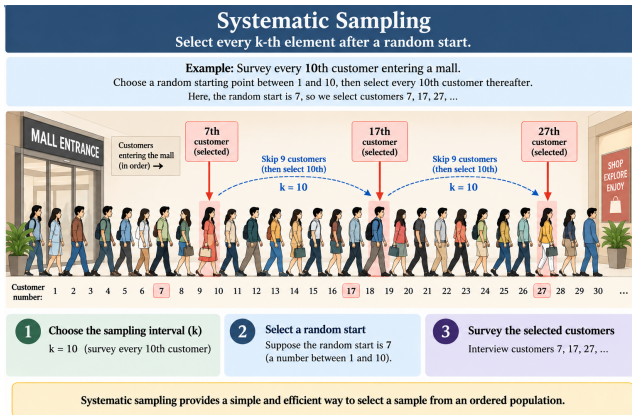


Figure: Systematic Sampling



Stratified Sampling

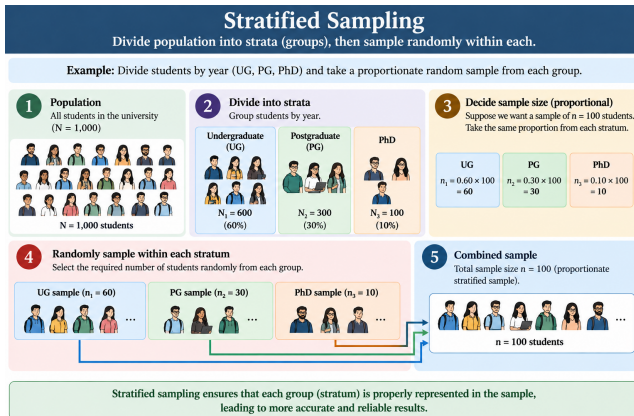


Figure: Stratified Sampling with Strata's of Different Size



Cluster Sampling

Cluster Sampling

Divide population into clusters (often geographically), then randomly choose some clusters and study all units inside them.

Example: Select 5 villages randomly and survey all households within them.

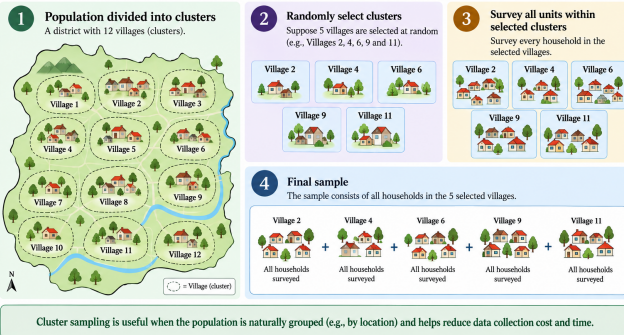


Figure: Cluster Sampling



Non-Probabilistic Sampling

Not all elements have a chance of being selected; depends on researcher's choice/judgment.

- **Convenience Sampling** : Choose whoever is easiest to reach.
For eg : Interviewing people outside your office.
- **Purposive (Judgmental) Sampling** : Researcher selects units based on specific purpose/criteria.
For eg : Selecting only expert farmers for a study.
- **Quota Sampling** : Similar to stratified, but selection within strata is non-random.
For eg : Interview 60 UG and 40 PG shoppers, chosen conveniently
- **Snowball Sampling** : Existing participants recruit future participants.
For eg : Studying rare diseases or hidden communities.



Convenience Sampling

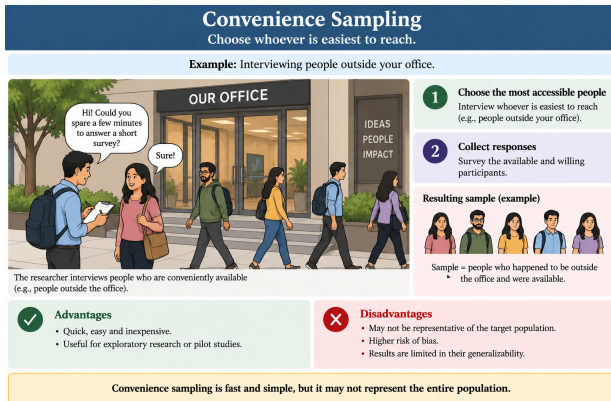


Figure: Convenience Sampling



Purposive or Judgemental Sampling

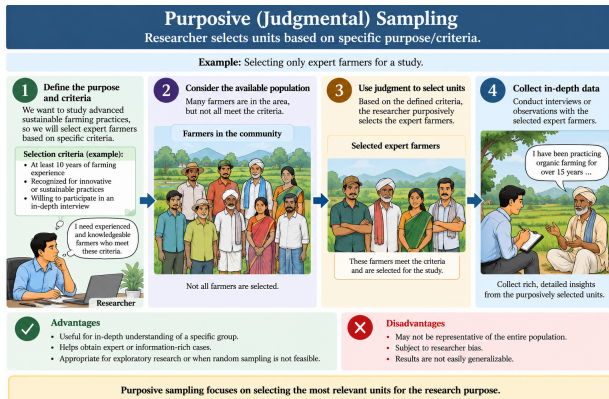


Figure: Purposive Sampling



Quota Sampling

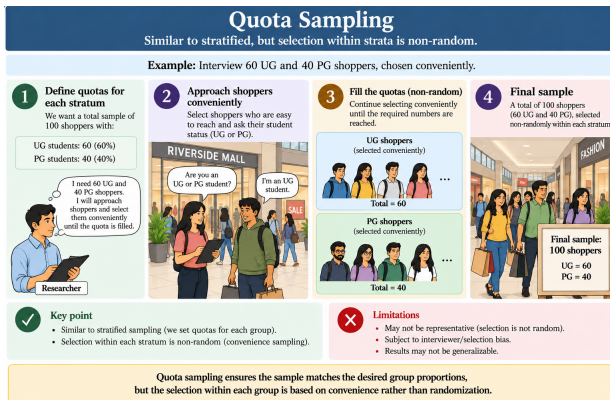


Figure: Quota Sampling



Snowball Sampling

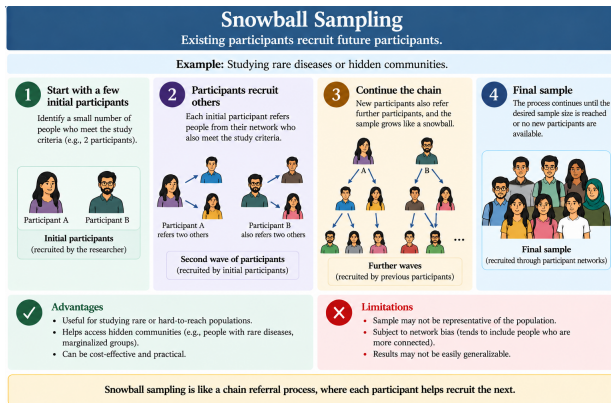


Figure: Snowball Sampling



Simple Random Sampling

Sampling Without Replacement

Suppose we select n units from a population of N units.
The number of different possible samples is

$$\binom{N}{n}$$

Using factorial notation,

$$\binom{N}{n} = \frac{N!}{n!(N-n)!}$$

In **Simple Random Sampling Without Replacement (SRSWOR)**, every possible sample has the same probability.
Therefore,

$$P(S = s) = \frac{1}{\binom{N}{n}}$$



Simple Random Sampling

Suppose there are five students:

$$U = \{A, B, C, D, E\}.$$

We want to randomly select $n=2$ students.

The number of possible samples is

$$\binom{5}{2} = \frac{5!}{2!3!} = 10.$$

The possible samples are

$$\begin{aligned} &\{A, B\}, \{A, C\}, \{A, D\}, \{A, E\}, \\ &\{B, C\}, \{B, D\}, \{B, E\}, \\ &\{C, D\}, \{C, E\}, \{D, E\}. \end{aligned}$$

Therefore,

$$P(\text{any particular sample}) = \frac{1}{10}$$



Simple Random Sampling

Inclusion Probability

We have

$$N = 5, \quad n = 2.$$

Consider student A .

Student A appears in four possible samples:

$$\{A, B\}, \quad \{A, C\}, \quad \{A, D\}, \quad \{A, E\}.$$

Therefore,

$$P(A \text{ is included}) = \frac{4}{10} = \frac{2}{5}$$

In general, under simple random sampling, probability that a unit is included in the sample is

$$\pi_i = P(i \in S) = \frac{n}{N}$$

for every population unit i .



Simple Random Sampling

Probability

1. Probability of a Particular Sample

$$P(S = s) = \frac{1}{\binom{N}{n}}$$

This refers to an entire sample of n units.

2. Inclusion Probability of a Particular Unit

$$\pi_i = P(i \in S) = \frac{n}{N}$$

This refers to whether one particular unit appears in the sample.

For $N = 5$ and $n = 2$:

$$P(\{A, B\}) = \frac{1}{10}, \quad P(A \in S) = \frac{2}{5}.$$



Simple Random Sampling

Inclusion Probability

Suppose we select n units from a population of N units.

The total number of possible samples of size n is

$$\binom{N}{n}$$

Now consider one particular unit i .

If unit i is included in the sample, we need to select the remaining $n - 1$ units from the other $N - 1$ units.

Therefore, the number of samples containing unit i is

$$\binom{N-1}{n-1}$$

Since every sample of size n is equally likely,

$$P(i \in S) = \frac{\binom{N-1}{n-1}}{\binom{N}{n}}.$$



Simple Random Sampling

Inclusion Probability

Expanding the combinations,

$$P(i \in S) = \frac{\frac{(N-1)!}{(n-1)!(N-n)!}}{\frac{N!}{n!(N-n)!}}$$

Hence,

$$P(i \in S) = \frac{(N-1)!}{(n-1)!(N-n)!} \frac{n!(N-n)!}{N!} = \boxed{\frac{n}{N}}.$$



Systematic Random Sampling

Inclusion Probability

N = Population size, n = Sample size

The sampling interval is

$$k = \frac{N}{n}$$

Select a random starting point

$$R \sim \text{Uniform}\{1, 2, \dots, k\}$$

Then select

$$R, R + k, R + 2k, \dots, R + (n - 1)k$$

For any population unit i , the inclusion probability is

$$\pi_i = P(i \text{ is selected}) = \frac{1}{k} = \frac{n}{N}$$



Stratified Random Sampling

Inclusion Probability

Divide the population into L strata:

$$N = N_1 + N_2 + \cdots + N_L$$

where N_h is the population size of stratum h .

From each stratum, select a simple random sample of size n_h :

$$n = n_1 + n_2 + \cdots + n_L$$

For an individual i belonging to stratum h , the **inclusion probability** is

$$\pi_{hi} = P(i \text{ is selected} \mid i \in h) = \frac{n_h}{N_h}$$



Stratified Random Sampling

Inclusion Probability

Under **proportionate stratified sampling**,

$$n_h = n \frac{N_h}{N}$$

Therefore,

$$\pi_{hi} = \frac{n_h}{N_h} = \frac{n}{N}$$

In proportionate stratified sampling, every individual has the same overall inclusion probability $\frac{n}{N}$.



Cluster Random Sampling

Inclusion Probability

Suppose the population is divided into M clusters:

$$C_1, C_2, \dots, C_M$$

Randomly select m clusters from the M clusters.

$$P(\text{Cluster } h \text{ is selected}) = \frac{m}{M}$$

In **one-stage cluster sampling**, all individuals inside a selected cluster are included in the sample.

Therefore, for an individual i belonging to cluster h ,

$$\pi_i = P(i \text{ is selected}) = P(C_h \text{ is selected}) = \frac{m}{M}$$



Parameter Estimation

Point Estimation, Interval Estimation & Hypothesis Testing

Parameter Estimation

- Parameter Estimation means estimating the parameters of the population distribution.
- Parameter Estimation has two methods
 - Point Estimation - Estimates the parameter as a single point or value. Examples are
 - Single point estimate for population mean is sample mean
 - Single point estimate for population variance is sample variance
 - Single point estimate for population proportion is sample proportion
 - Interval Estimation - Estimates the interval with lower and upper limits for a parameter. Examples are
 - Interval Estimates for population mean with a upper and lower limit
 - Interval Estimates for population variance with a upper and lower limit
 - Interval Estimates for population proportion with a upper and lower limit



Point Estimation

Unbiased, Consistent and Efficient Estimators

Statistic

Statistic for One Groups

Definition

A **statistic** is a numerical value calculated from sample data that is used to describe or summarize some characteristic of the sample.

Formally, if a random sample is $X_1, X_2, \dots, X_n$, then a statistic is any function of the sample observations:

$$T = f(X_1, X_2, \dots, X_n)$$

- Sample Mean: $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
- Sample Variance: $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
- Sample Proportion: $\hat{p} = \frac{k}{n}$, where k is the number of successes, n is the sample size.

A **Parameter** is a numerical characteristic of the population (e.g., μ, σ^2, p) and a **Statistic** is a numerical characteristic of the sample (e.g., $\bar{x}, s^2, \hat{p}$).



- Sample Difference of Means:

$$\bar{X}_1 - \bar{X}_2 = \frac{1}{n_1} \sum_{i=1}^{n_1} X_{1i} - \frac{1}{n_2} \sum_{i=1}^{n_2} X_{2i}$$

- Sample Difference of Proportions:

$$\hat{p}_1 - \hat{p}_2 = \frac{k_1}{n_1} - \frac{k_2}{n_2},$$

where k_1 and k_2 are the number of successes in samples of size n_1 and n_2 , respectively.



Point Estimation

Definition

A **point estimator** is a statistic used to provide a **single best estimate** of an unknown population parameter. The value obtained from the sample is called a **point estimate**.

Population Parameter	Point Estimator (Sample Statistic)
Mean (μ)	Sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
Variance (σ^2)	Sample variance $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
Proportion (p)	Sample proportion $\hat{p} = \frac{k}{n}$



Point Estimation

- A point estimator is a **random variable** because it depends on the sample.
- Desirable properties:
 - **Unbiasedness:** Expected value equals the population parameter
 - **Consistency:** Estimate approaches true parameter as sample size increases
 - **Efficiency:** Smaller variance among unbiased estimators



Unbiased Estimator

Definition

An estimator $\hat{\theta}$ of a population parameter θ is said to be **unbiased** if its expected value equals the true parameter:

$$E[\hat{\theta}] = \theta$$

If $E[\hat{\theta}] \neq \theta$, the estimator is **biased**, and the difference is called **bias**:

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

Parameter	Estimator	Unbiased?
Mean (μ)	Sample mean $\bar{X}$	Yes, $E[\bar{X}] = \mu$
Variance (σ^2)	Sample variance s^2	Yes, $E[s^2] = \sigma^2$
Proportion (p)	Sample proportion $\hat{p}$	Yes, $E[\hat{p}] = p$



Consistent Estimator

Definition

An estimator $\hat{\theta}_n$ of a population parameter θ is said to be **consistent** if it converges in probability to the true parameter as the sample size $n \rightarrow \infty$:

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty$$

Intuitively: as the sample size increases, the estimator gets **closer and closer** to the true parameter.

Note : There are biased and unbiased estimators that are consistent.



Efficient Estimator

Definition

An estimator $\hat{\theta}$ of a population parameter θ is said to be **efficient** if it is **unbiased** and has the **smallest variance** among all unbiased estimators of θ :

$$\text{Var}(\hat{\theta}) \leq \text{Var}(\tilde{\theta}) \quad \text{for any other unbiased estimator } \tilde{\theta}.$$

Efficiency measures how **precisely** an estimator estimates the parameter.

- For a normal population $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, the **sample mean** $\bar{X}$ is an efficient estimator of μ .
- Among all unbiased estimators of μ , $\bar{X}$ has the minimum variance σ^2/n .



Efficient Estimator

Relative Efficiency

$$\text{Relative Efficiency of } \hat{\theta}_1 \text{ to } \hat{\theta}_2 = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

Step 1: Identify estimators

$$\hat{\theta}_1 = \text{sample mean } \bar{X}, \quad \hat{\theta}_2 = \text{sample median}$$

Step 2: Variances for normal distribution

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \text{Var}(\text{median}) \approx \frac{\pi\sigma^2}{2n} \quad (\text{for large } n)$$

Step 3: Compute relative efficiency

$$\text{RE}(\text{mean, median}) = \frac{\text{Var}(\text{median})}{\text{Var}(\bar{X})} = \frac{\frac{\pi\sigma^2}{2n}}{\frac{\sigma^2}{n}} = \frac{\pi}{2} \approx 1.57$$

Relative efficiency of 1.57 means the sample mean achieves the same precision with fewer observations than the sample median.



Interval Estimation

Interval Estimation

Definition (Interval Estimation)

Interval estimation refers to estimating a population parameter (like a mean or proportion or variance) using a range of plausible values instead of a single point estimate.

Confidence Interval (CI) = Point Estimate (PE) $\pm$ Margin of Error (ME)

The confidence interval of a population parameter is an interval between PE-ME to PE+ME.



Interval Estimation

Motivation

- Assume the sample mean is $\hat{\mu} = 1000$
- The sample mean does not say anything about how close the sample mean $\hat{\mu}$ is to the population mean μ
- Is the population mean likely to be in the range 900 and 1000 or $900 \leq \mu \leq 1000$
- Is the population mean likely to be in the range 990 to 1010 or $990 \leq \mu \leq 1010$?
- An interval estimate for a population parameter is called a **confidence interval**.
- Shorter the length of the interval, more precise is the estimation.



Interval Estimation

CONFIDENCE INTERVAL ON THE MEAN OF A NORMAL DISTRIBUTION, VARIANCE KNOWN

- Suppose that $X_1, X_2, \dots, X_n$ is a random sample from a normal distribution with unknown mean μ and known variance σ^2 .
- CLT says

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \quad (1)$$

As $n \rightarrow \infty$, Z is the standard normal distribution.

- A confidence interval estimate for μ is an interval of the form $l \leq \mu \leq u$, where the end points l and u are computed from the sample data.
- Because different samples will produce different values of l and u , these end-points are values of random variables L and U , respectively.



Interval Estimation

CONFIDENCE INTERVAL ON THE MEAN OF A NORMAL DISTRIBUTION, VARIANCE KNOWN

- Is it possible to determine the values of L and U such that the following probability condition is satisfied. $P[L \leq \mu \leq U] = 1 - \alpha$, where $1 - \alpha$ is called the confidence.
- The answer is yes! $P[-z_{\frac{\alpha}{2}} \leq Z \leq z_{\frac{\alpha}{2}}] = 1 - \alpha$
where $Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$
- For a sample of size n, the confidence interval for population mean μ and population standard deviation σ with a confidence of $1 - \alpha$, where α is the level of significance, is given by

$$P\left[\bar{X} - ME \leq \mu \leq \bar{X} + ME\right] = 1 - \alpha \quad (2)$$

$$ME = z_{\frac{\alpha}{2}} \times SE$$

$$SE = \frac{\sigma}{\sqrt{n}}$$



Interval Estimation

CONFIDENCE INTERVAL ON THE MEAN OF A NORMAL DISTRIBUTION, VARIANCE KNOWN

$$P\left[\bar{X} - z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha$$

The confidence interval for population mean μ is defined as $\left[\bar{X} \pm z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}\right]$.

The width of the confidence interval represents the precision of the estimate.

- The width of the confidence interval (Upper Limit - Lower Limit) is given by $2ME$ or $2 \cdot z_{\frac{\alpha}{2}} * SE$ or $2 \cdot z_{\frac{\alpha}{2}} * \frac{\sigma}{\sqrt{n}}$
- When confidence $(1-\alpha)$ is increased, $Z_{\frac{\alpha}{2}}$ is increased. For example : For 95% confidence or $\alpha = .05$, we have $Z_{.025} = 1.96$ and 99% confidence or $\alpha = .1$ or $Z_{.005} = 2.58$.
- Confidence level $\uparrow$ Confidence Interval $\uparrow$ Precision $\downarrow$



Interval Estimation

CONFIDENCE INTERVAL ON THE MEAN OF A NORMAL DISTRIBUTION, VARIANCE KNOWN

For population mean $\mu = 50$, $\sigma = 10$ and $n = 30$, $z_{.025} = 1.96$, the 95% confidence intervals are shown below.

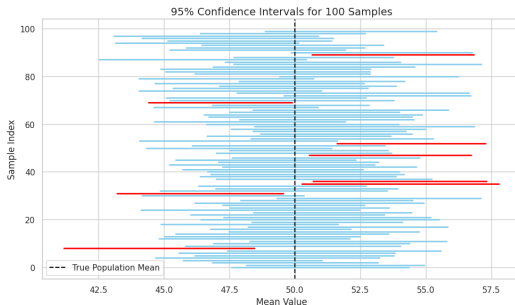


Figure: Confidence Interval for Population Mean



Interval Estimation

Precision and Confidence Interval

- Ideally you would want a high precision and high confidence interval.
- But when we increase the confidence level or the confidence interval, the precision is actually decreasing.
- **But for a given confidence interval, we can improve the precision by increasing the sample size n**
- The sampling error is the difference between a sample statistic and the true population parameter it estimates. Example $\bar{X} - \mu$ or $\hat{p} - p$



Interval Estimation

Choice of Sample Size and Error

$$P\left[\bar{X} - z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha$$

$$P\left[-z_{\frac{\alpha}{2}} \leq \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq z_{\frac{\alpha}{2}}\right] = 1 - \alpha$$

$$P\left[|E| < z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha$$

$$P\left[|E| < E_{max}\right] = 1 - \alpha$$

If $\bar{X}$ is the sample mean and μ is the population mean, we can be $100(1 - \alpha)$ sure that the sampling error $\bar{X} - \mu$ will not exceed an error

E_{max} if the sample size $n = \left[\frac{z_{\frac{\alpha}{2}} * \sigma}{E_{max}}\right]^2$



Interval Estimation

Error and Sample Size

- Error $E_{max} \downarrow$ implies increase in the sample size n if the confidence level $(1 - \alpha)$ and population standard deviation σ is fixed.
- As population standard deviation $\sigma \downarrow$ the sample size n also reduces when the error E and the confidence level $(1 - \alpha)$ is fixed.
- As the level of confidence increases or $z_{\frac{\alpha}{2}}$ increases, the sample size n also increases when the error E and population standard deviation σ is fixed.

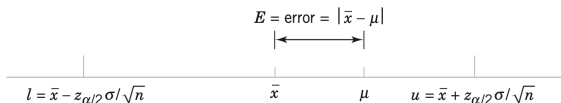


Figure: Error in estimating μ with $\bar{X}$



Interval Estimation

CONFIDENCE INTERVAL ON THE MEAN OF A NORMAL DISTRIBUTION, VARIANCE UNKNOWN - Large Sample ($n > 40$)

- Suppose that $X_1, X_2, \dots, X_n$ is a random sample from a normal distribution with unknown mean μ and unknown variance σ^2 .
- CLT says

$$Z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

As $n \rightarrow \infty$, Z is the standard normal distribution.

$$P\left[\hat{\mu} - z_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}} \leq \mu \leq \hat{\mu} + z_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}\right] = 1 - \alpha$$



Interval Estimation

CONFIDENCE INTERVAL ON THE MEAN OF A NORMAL DISTRIBUTION, VARIANCE UNKNOWN - Small Sample ($n \leq 40$)

- Suppose that $X_1, X_2, \dots, X_n$ is a random sample from a normal distribution with unknown mean μ and unknown variance σ^2 .
- CLT says

$$T = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

As $n \rightarrow \infty$, is the T distribution with $n-1$ degrees of freedom

$$P\left[\hat{\mu} - t_{\frac{\alpha}{2}, n-1} \cdot \frac{s}{\sqrt{n}} \leq \mu \leq \hat{\mu} + t_{\frac{\alpha}{2}, n-1} \cdot \frac{s}{\sqrt{n}}\right] = 1 - \alpha$$



Interval Estimation

Large Sample Confidence Interval for a Population Proportion

- We need to construct confidence interval for population proportion.
- Assume we randomly sample $x_1, x_2, x_3, \dots, x_n$
- Note that the samples $x_1, x_2, x_3, \dots, x_n$ are drawn from a bernouli distribution of parameter p .
- The sample proportion is the point estimator of unknown population population proportion p .

$$\hat{p} = \frac{\sum_{i=1}^n [x_i]}{n}$$

- $\hat{p}$ is a random variable with mean $\frac{1}{n} * n * p = p$ and variance $\frac{1}{n^2} * n * p * (1 - p) = \frac{p(1-p)}{n}$.
- As per CLT, if n is large, the distribution of Z approaches $\mathcal{N}(0, 1)$

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p \cdot (1-p)}{n}}} \quad (3)$$



Interval Estimation

Large Sample Confidence Interval for a Population Proportion



$$P\left[-z_{\frac{\alpha}{2}} < Z < z_{\frac{\alpha}{2}}\right] = 1 - \alpha$$



$$P\left[\hat{p} - z_{\frac{\alpha}{2}} \sqrt{\frac{p \cdot (1 - p)}{n}} \leq p < \hat{p} + z_{\frac{\alpha}{2}} \sqrt{\frac{p \cdot (1 - p)}{n}}\right] = 1 - \alpha$$

- Confidence Interval for Proportion

$$P\left[\hat{p} - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}} \leq p < \hat{p} + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}}\right] = 1 - \alpha$$



Interval Estimation

Large Sample Confidence Interval for a Population Proportion

- Choice of sample size n based on max error E_{max} or $\max|(\hat{p} - p)|$

$$P\left[|E| < E_{max}\right] = 1 - \alpha$$

If $\hat{p}$ is the sample proportion and p is the population proportion, we can be $100(1 - \alpha)\%$, that the error $|\hat{p} - p|$ will not exceed an error

E_{max} if the sample size $n = \left[\frac{z_{\frac{\alpha}{2}} * \sigma}{E_{max}}\right]^2$ or $\left[\frac{z_{\frac{\alpha}{2}} * \sqrt{p \cdot (1-p)}}{E_{max}}\right]^2$



Interval Estimation

Large Sample Confidence Interval for a Population Proportion

Out of random sample of 85 students, 10 students voted for party A and 75 voted for the rest. Compute the confidence interval for the population proportion voting for party A

Sample proportion $\hat{p} = \frac{10}{85}$ For a two sided 95% interval, $\alpha = .05$, $z_{.025} = 1.96$

The confidence interval of population proportion is given by

$$\left[\hat{p} - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}} \leq p < \hat{p} + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}} \right]$$

$$.05 \leq p \leq .19$$

How many samples do we pick for taking a decision assuming the max error $\max |(\hat{p} - p)|$ is .05?

$$n = \left[\frac{z_{\frac{\alpha}{2}} * \sqrt{p \cdot (1-p)}}{E_{max}} \right]^2$$

$$\text{For observed sample, sample size } n = \left[\frac{z_{\frac{\alpha}{2}} * \sqrt{\hat{p} \cdot (1-\hat{p})}}{E_{max}} \right]^2 = 163$$

$$\text{If no observed sample is there, maximum sample size } n = \left[\frac{z_{\frac{\alpha}{2}} * \sqrt{.5 * (1-.5)}}{E_{max}} \right]^2 =$$

385



Interval Estimation

Large Sample Confidence Interval for a Population Proportion

In the Framingham Offspring study 1,219 subjects were on anti-hypertensive medication out of 3,532 total subjects. Construct the confidence interval on the population proportion with 95% confidence?

- $\hat{p} = \frac{1219}{3532}$
- Confidence interval equals to $[\hat{p} - E, \hat{p} + E]$
- Error E is $z_{\frac{\alpha}{2}} * SE(\hat{p})$
- $SE(\hat{p}) = \sqrt{\frac{p \cdot (1-p)}{n}}$
- Since p is unknown, $SE(\hat{p}) = \sqrt{\frac{\hat{p} \cdot (1-\hat{p})}{n}}$
- n is sample size = 3532
- Confidence Interval is [.329, .361]



Interval Estimation

Confidence Interval for Population Variance

- If $X_1, X_2, X_3, \dots, X_n$ are standard normal variables with mean 0 and variance 1, the variable χ^2 expressed as the sum of squares of standard normal variable would have a χ^2 distribution.

$$\chi^2 = X_1^2 + X_2^2 + X_3^2 + \dots + X_n^2 \quad (4)$$

- $\chi_{n-1}^2 = \frac{n-1 \cdot s^2}{\sigma^2}$ is a chi-square distribution with n-1 degrees of freedom.

- $P\left[\chi_{1-\frac{\alpha}{2}, n-1}^2 \leq \chi^2 \leq \chi_{\frac{\alpha}{2}, n-1}^2\right] = 1 - \alpha$

- $\chi_{1-\frac{\alpha}{2}, n-1}^2 \leq \frac{n-1 \cdot s^2}{\sigma^2} \leq \chi_{\frac{\alpha}{2}, n-1}^2$

- The confidence interval of σ^2 is $\left[\frac{(n-1)s^2}{\chi_{\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}\right]$



Interval Estimation - One Group

Confidence Intervals for Population Mean, Proportion, Variance

Parameter	Interval Type	Distribution Used
Mean (μ)	$\bar{X} \pm z_{\alpha} \frac{s}{\sqrt{n}}$	t -distribution , $n > 40$
Mean (μ)	$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$	t -distribution , $n < 40$
Proportion (p)	$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	Normal (z)
Variance (σ^2)	$\frac{(n-1)s^2}{\chi^2_{\alpha/2, n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2, n-1}}$	Chi-square



Confidence Interval and Hypothesis Testing

Means Test

A nutritionist measures the daily vitamin C intake of a random sample of $n = 25$ adults. The sample mean is $\bar{x} = 100$ mg and the sample standard deviation is $s = 15$ mg.

- 1 Construct a 95% confidence interval for the population mean μ and decide whether there is evidence at the 5% level that the true mean differs from $\mu_0 = 95$ mg.

$$n = 25, \quad \bar{x} = 100, \quad s = 15, \quad \text{confidence level} = 95\%$$

$$df = n - 1 = 24, \quad t_{0.025, 24} \approx 2.064$$

$$SE = \frac{s}{\sqrt{n}} = \frac{15}{\sqrt{25}} = 3$$

$$ME = t_{0.025, 24} \cdot SE = 2.064 \times 3 \approx 6.192$$

$$\bar{x} \pm ME = 100 \pm 6.192 \Rightarrow (93.808, 106.192)$$



Confidence Interval and Hypothesis Testing

Proportions Test

- A public-health survey of $n = 500$ adults found that $x = 65$ reported using Drug X during the past month.
- Construct a 95% confidence interval for the population proportion p of adults who use Drug X and check whether the null hypothesis $H_0 : p = .12$ can be rejected.

$$\hat{p} = \frac{x}{n} = \frac{65}{500} = 0.13$$

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0.13 \times 0.87}{500}} \approx 0.01504$$

$$ME = z_{\alpha/2} \times SE(\hat{p}) = 1.96 \times 0.01504 = 0.02948$$

$$\hat{p} \pm ME = 0.13 \pm 0.02948$$

$$\Rightarrow (0.1005, 0.1595)$$



Hypothesis Testing - A Motivation

Rocket Fuel Burn Rate Case

Hypothesis Testing - A motivation

Suppose that an engineer is designing an air crew escape system that consists of an ejection seat and a rocket motor that powers the seat. The rocket motor contains a propellant, and in order for the ejection seat to function properly, the propellant should have a mean burning rate of 50 cm/sec. If the burning rate is too low, the ejection seat may not function properly, leading to an unsafe ejection and possible injury of the pilot. Higher burning rates may imply instability in the propellant or an ejection seat that is too powerful, again leading to possible pilot injury.

The important question is **"Is the mean burning rate equal to 50 cm/sec"**.

The solution is **Hypothesis Testing**



Hypothesis Testing

Definitions

- The statements made about the parameters of the general population are called as hypotheses.
- **No hypothesis is made about samples.**
- The decision making procedure is called hypothesis testing
- A **statistical hypothesis** is a statement about the parameters of one or more populations.
- Since we use probability distributions to represent populations, a statistical hypothesis may also be thought of as a statement about the probability distribution of a random variable. The hypothesis will usually involve one or more parameters of this distribution.
- $H_0 : \mu = 50$
 $H_1 : \mu \neq 50$
- H_1 is double sided. $H_1 : \mu \neq 50$ means $\mu > 50$ or $\mu < 50$



Test of Statistical Hypothesis

- Hypotheses -

Null Hypothesis H_0 and Alternate Hypothesis H_1

$$H_0 : \mu = 50 \text{ cm/sec}$$

$$H_1 : \mu \neq 50 \text{ cm/sec}$$

- Take n samples $X_1, X_2 \dots X_n$ of the fuel and perform the fuel burn test and find the average burning rate $\bar{X}$
- If the average burning rate $\bar{X}$ is close to the real unknown population mean μ , accept the Null Hypothesis. Else fail to accept the Null Hypothesis.



Acceptance & Rejection Region

- Suppose if the sample mean $\bar{X}$ is between 48.5 cm/sec and 51.5 cm/sec or $48.5 \leq \bar{X} \leq 51.5$ and we decide to accept the null hypothesis $H_0 : \mu = 50$ cm/sec.
- Suppose if the sample mean $\bar{X}$ is lesser than 48.5 or greater than 51.5, we decide to not accept the null hypothesis $H_0 : \mu = 50$ cm/sec or accept the alternate hypothesis $H_1 : \mu \neq 50$ cm/sec

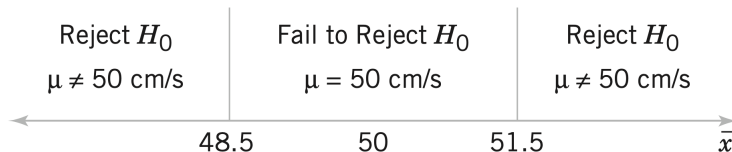


Figure: Critical Regions & Acceptance Region, Critical Values



Hypothesis Testing & Wrong Conclusions : Type I and Type II Errors

- Scenario 1 : There is a chance that the true mean μ was indeed 50 cm/sec. However, due to some sampling issue, the sample mean $\bar{X}$ was less than 48.5 or greater than 51.5. The chance or the probability that we fail to accept the null hypothesis given that the null hypothesis is true is called as the significance level α .
 $\alpha = P(\text{Type I Error}) = P(\text{Reject } H_0 \text{ when } H_0 \text{ is true})$
- Rejecting the null hypothesis H_0 when it is true is defined as a type I error.
- Scenario 2 : The true mean is not 50 cm/sec. However, due to some sampling issue, the sample mean $\bar{X}$ was found to be between 48.5 and 51.5. The chance or the probability that we accept the null hypothesis given that the null hypothesis is false is called as β error.
- Failing to reject the null hypothesis when it is false is defined as a type II error. $\beta = P(\text{Type II Error}) = P(\text{Fail to Reject } H_0 \text{ when } H_0 \text{ is false})$



Computing Type 1 Error Probability or Significance Level α

- $\alpha = P(\bar{X} < 48.5/\mu = 50) + P(\bar{X} > 51.5/\mu = 50)$
- Assume population mean $\mu = 50$ and standard deviation of burning rate $\sigma = 2.5$ and number of samples = 10, the sample mean $\bar{X}$ would have a mean $\mu = 50$ and standard deviation of $\frac{\sigma}{\sqrt{n}}$ or .79

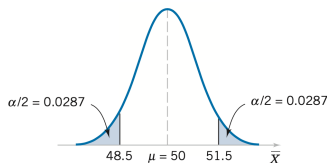


Figure: Type 1 Error

- $\alpha = P(Z < -1.9) + P(Z > 1.9) = .0287 + .0287 = .0574$
- This means that that 5.74% of all random samples would lead to rejection of the null hypothesis H_0 even if the null hypothesis is true
- Assume you repeat the experiment of taking n samples $X_1, X_2 \dots X_n$ 100 times, around 5 times, you might get a rejection of null hypothesis even if the null hypothesis was really true.



Type 1 Error and Sample Size n

- If you increase the sample size, it is reasonable to expect that the type 1 error is reduced.
- Assume the sample size is increased from 10 to 16, standard deviation of $\frac{\sigma}{\sqrt{n}}$ would be .625.
- $\alpha = P(Z < -2.4) + P(Z > 2.4) = .0082 + .0082 = .0164$
- From 5.74%, we have reduced the Type 1 Error to 1.64%.



Computing Type 2 Error Probability

- The null hypothesis is false for Type 2 Error or $\mu \neq 50$. It means μ can be 51 or 52 or 49 or 48 or anything other than 50.
- The type 2 error probability can be computed only for some specific values of population mean μ
- We can compute the type 2 error probability of $\mu = 48$ and $\mu = 52$.
- In short, we are trying to compute the probability of accepting the null hypothesis $H_0 : \mu = 50$ when the true mean μ is 52.
- $\beta = P(48.5 \leq \bar{X} \leq 51.5)$ when true mean μ is 52.
- For sample mean $\bar{X}$ with mean 52 and standard deviation $\frac{\sigma}{\sqrt{n}}$, where $\sigma = 2.5$
- $\beta = P(-4.43 \leq Z \leq -.63) = .2643 - .00000 = .2643$
- Thus, if we are testing $H_0: \mu = 50$ against $H_1: \mu \neq 50$ with n samples $n = 10$, and the true value of the mean $\mu = 52$, the probability that we will fail to reject the false null hypothesis ($\mu = 50$) is 0.2643.
- Assume you repeat the experiment of taking n samples $X_1, X_2 \dots X_n$ 100 times, around 26 times, you might get a acceptance of null hypothesis even if the null hypothesis was wrong.



Computing Type 2 Error Probability

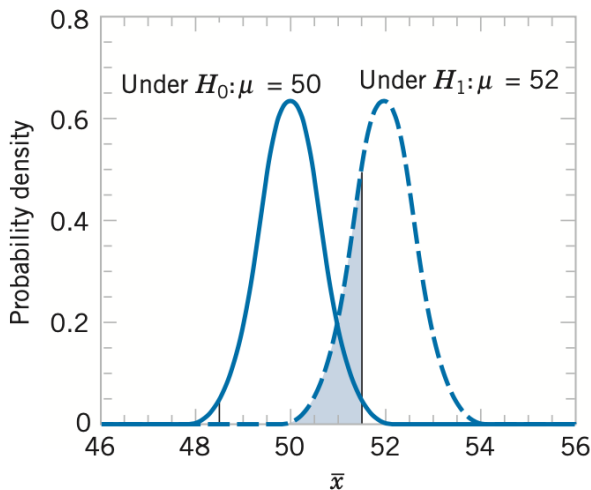


Figure: Type 2 Error [1]



Decisions in Hypothesis Testing - Summary

Decision	H_0 is True	H_0 is False
Fail to reject H_0	No error	Type II error
Reject H_0	Type I error	No error



Hypothesis Testing and Power of Statistical Test

- Type 1 and Type 2 errors are related. Both cannot be simultaneously reduced.
- The general idea is to fix the significance level α and increase the sample size n to reduce the beta error β , provided that the α is a constant.
- When the null hypothesis is false, it has to be rejected.
- The probability of rejecting a false null hypothesis is called the power of a test
- The power is the probability of rejecting a false null hypothesis
- Power of a test = $1 - \beta$, where β is the beta error or the probability of failing to reject H_0 when H_0 is false.
- When μ is 52, the power of the statistical test is $1 - .2643 = .7357$.



Hypothesis Testing

Test for Mean, Proportion and Variance
Single Group | Two Group | Multiple Groups

Hypothesis Testing

Hypothesis Testing

A hypothesis is an assumption or statement about a population parameter (such as mean, variance, or proportion) that we test using sample data.

- Groups - Single Group | Two Group | Multiple Groups
- Samples - Dependent (Paired) Samples | Independent Samples
- Assumptions - Parameteric | Non-parametric Tests



Hypothesis Tesing

- Hypothesis for Population Mean - One Group
- Hypothesis for Population Mean - Two Group
- Hypothesis for Population Mean - Multiple Group
- Hypothesis for Population Proportion - One Group
- Hypothesis for Population Proportion - Two Group
- Hypothesis for Population Proportion - Multiple Group
- Hypothesis for Population Variance - One Group
- Hypothesis for Population Variance - Two Group
- Hypothesis for Population Variance - Multiple Group



Hypothesis for Population Mean

Z-Test or T-Test

One-Mean Hypothesis

A hypothesis is an assumption or statement about a population mean that we test using sample data.

- **Null Hypothesis** (H_0): The default assumption, states no effect or no difference.

$$H_0 : \mu = \mu_0$$

- **Alternative Hypothesis** (H_1): Competes with H_0 , states that an effect or difference exists.

Two-tailed: $H_1 : \mu \neq \mu_0$

Right-tailed: $H_1 : \mu > \mu_0$

Left-tailed: $H_1 : \mu < \mu_0$



Hypothesis for Population Proportion

Z-Test

One-Proportion Hypothesis

A hypothesis is an assumption or statement about a population proportion that we test using sample data.

- **Null Hypothesis** (H_0): The default assumption, states no effect or no difference.

$$H_0 : p = p_0$$

- **Alternative Hypothesis** (H_1): Competes with H_0 , states that an effect or difference exists.

Two-tailed: $H_1 : p \neq p_0$

Right-tailed: $H_1 : p > p_0$

Left-tailed: $H_1 : p < p_0$



Hypothesis for Population Variance

Chi-Square Test

One-Variance Hypothesis

A hypothesis is an assumption or statement about a population variance that we test using sample data.

- **Null Hypothesis** (H_0): The default assumption, states no effect or no difference.

$$H_0 : \sigma^2 = \sigma_0^2$$

- **Alternative Hypothesis** (H_1): Competes with H_0 , states that an effect or difference exists.

$$\text{Two-tailed: } H_1 : \sigma^2 \neq \sigma_0^2$$

$$\text{Right-tailed: } H_1 : \sigma^2 > \sigma_0^2$$

$$\text{Left-tailed: } H_1 : \sigma^2 < \sigma_0^2$$



Hypothesis for Two Population Means

Independent and Dependent Sample Tests

Two-Mean Hypothesis

A hypothesis is an assumption or statement about the difference between two population means that we test using sample data.

- **Null Hypothesis** (H_0): No difference between the two population means.

$$H_0 : \mu_1 = \mu_2 \quad \text{or} \quad H_0 : \mu_1 - \mu_2 = 0$$

- **Alternative Hypothesis** (H_1): A difference exists between the two population means.

$$\text{Two-tailed: } H_1 : \mu_1 \neq \mu_2$$

$$\text{Right-tailed: } H_1 : \mu_1 > \mu_2$$

$$\text{Left-tailed: } H_1 : \mu_1 < \mu_2$$



Hypothesis for Two Population Proportions

Z-Test

Two-Proportion Hypothesis

A hypothesis is an assumption or statement about the difference between two population proportions that we test using sample data.

- **Null Hypothesis** (H_0): No difference between the two population proportions.

$$H_0 : p_1 = p_2 \quad \text{or} \quad H_0 : p_1 - p_2 = 0$$

- **Alternative Hypothesis** (H_1): A difference exists between the two population proportions.

$$\text{Two-tailed: } H_1 : p_1 \neq p_2$$

$$\text{Right-tailed: } H_1 : p_1 > p_2$$

$$\text{Left-tailed: } H_1 : p_1 < p_2$$



Hypothesis for Two Population Variances

F-Test

Two-Variance Hypothesis

A hypothesis is an assumption or statement about the ratio of two population variances that we test using sample data.

- **Null Hypothesis** (H_0): The two variances are equal.

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad \text{or} \quad H_0 : \frac{\sigma_1^2}{\sigma_2^2} = 1$$

- **Alternative Hypothesis** (H_1): The variances differ.

$$\text{Two-tailed: } H_1 : \sigma_1^2 \neq \sigma_2^2$$

$$\text{Right-tailed: } H_1 : \sigma_1^2 > \sigma_2^2$$

$$\text{Left-tailed: } H_1 : \sigma_1^2 < \sigma_2^2$$



Hypothesis for Multiple Population Means

Analysis of Variance (ANOVA)

Multiple-Mean Hypothesis

A hypothesis is an assumption or statement about the means of multiple populations that we test using sample data.

- **Null Hypothesis** (H_0): All population means are equal.

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

- **Alternative Hypothesis** (H_1): At least one population mean is different.

$$H_1 : \text{Not all } \mu_i \text{ are equal, } i = 1, \dots, k$$



Hypothesis for Multiple Population Proportions

Chi-square Test for Independence

Multiple-Proportion Hypothesis

A hypothesis is an assumption or statement about the proportions of multiple populations that we test using sample data.

- **Null Hypothesis** (H_0): All population proportions are equal.

$$H_0 : p_1 = p_2 = \cdots = p_k$$

- **Alternative Hypothesis** (H_1): At least one population proportion is different.

$$H_1 : \text{Not all } p_i \text{ are equal, } i = 1, \dots, k$$



Hypothesis for Multiple Population Variances

Bartlett's Test, Levenes Test

Multiple-Variance Hypothesis

A hypothesis is an assumption or statement about the variances of multiple populations that we test using sample data.

- **Null Hypothesis** (H_0): All population variances are equal.

$$H_0 : \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$$

- **Alternative Hypothesis** (H_1): At least one population variance is different.

$$H_1 : \text{Not all } \sigma_i^2 \text{ are equal, } i = 1, \dots, k$$



Hypothesis Testing Framework

Step 1: State the Hypotheses

- Two-tailed test:

$$H_0 : \theta = \theta_0, \quad H_1 : \theta \neq \theta_0$$

- Right-tailed test:

$$H_0 : \theta = \theta_0, \quad H_1 : \theta > \theta_0$$

- Left-tailed test:

$$H_0 : \theta = \theta_0, \quad H_1 : \theta < \theta_0$$

Step 2: Choose Significance Level (α)

- Commonly $\alpha = 0.05$ or 0.01
- α is the probability of Type I error (rejecting H_0 when true)



Hypothesis Testing Framework

Step 3: Select Test Statistic

Situation	Test Statistic
Population mean, known σ , large samples	$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$
Population mean, unknown σ , small samples	$t = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$
Population proportion	$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$
Population variance	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$
Comparing two variances	$F = \frac{s_1^2}{s_2^2}$



Hypothesis Testing Framework

Step 4: Determine Rejection Region

- Two-tailed:

$$|Z| > Z_{\alpha/2}, \quad t > t_{\alpha/2} \text{ or } t < -t_{\alpha/2}, \quad \chi^2 < \chi_{\alpha/2, df}^2 \text{ or } \chi^2 > \chi_{1-\alpha/2, df}^2$$

- Right-tailed:

$$Z > Z_{\alpha}, \quad t > t_{\alpha}, \quad \chi^2 > \chi_{\alpha, df}^2$$

- Left-tailed:

$$Z < -Z_{\alpha}, \quad t < -t_{\alpha}, \quad \chi^2 < \chi_{1-\alpha, df}^2$$

Step 5: Make Decision

- Reject H_0 if test statistic falls in rejection region
- Otherwise, fail to reject H_0



Statistical Tests

One Group, Two Group and Multi Groups

Test for Means

One Group & Two Group & Multi-Group Tests

One Group Tests

Z & t Tests

Test for Means



Abraham De Moivre
Z-Distribution



Willain Sealy Gosset
t-Distribution



z-Distribution

- The **z-distribution** is the **standard normal distribution**, with mean 0 and standard deviation 1:

$$Z \sim N(0, 1)$$

- Any normal variable $X \sim N(\mu, \sigma^2)$ can be **standardized** to a z-score using:

$$Z = \frac{X - \mu}{\sigma}$$

- The z-distribution is symmetric about 0, and the total area under the curve is 1.
- Abraham de Moivre first introduced the normal distribution while studying the binomial distribution.
- When the population variance σ^2 is known, the z-test statistic for a sample mean $\bar{X}$ is:

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$



t-Distribution

- William Gosset (Student) showed that when the sample size is small, the t-statistic for a sample mean $\bar{X}$ is

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

follows a **t-distribution** rather than a normal distribution.

- The **t-distribution** is wider and has heavier tails than the normal distribution, reflecting the increased uncertainty in $\bar{X}$ due to small n .
- The shape of the t-distribution depends on the degrees of freedom:

$$df = n - 1$$

As n increases, the t-distribution approaches the normal distribution. Therefore, for large samples ($n \geq 30$), a z-test can be safely used.



Z-distribution and t-distribution

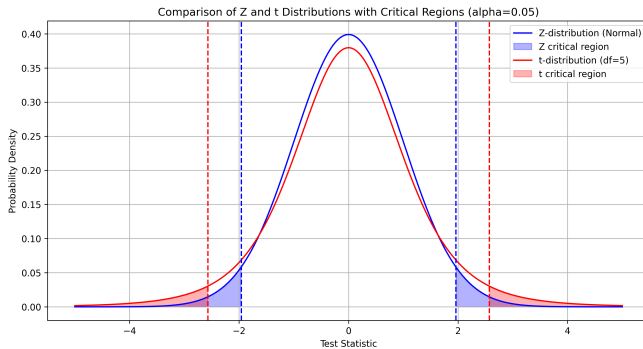


Figure: Z-distribution and t-distribution



Test for Means - One Group

- Test for Means - One Group is a parametric test as the following assumptions are made.
 - Normality : The population has a normal distribution.
 - Scale : Data should be measured at interval or ratio scale.
 - Random Sampling : Samples are randomly drawn.
 - Independent Observations: Samples are independent of each other.
- If sample size $n < 30$, use a t-test if " $X_1, X_2, \dots, X_n \sim$ i.i.d. with mean μ , unknown σ , and approximately normal population".
- If sample size $n > 30$, use a Z-test if " $X_1, X_2, \dots, X_n \sim$ i.i.d. with mean μ , known/unknown σ ."



Test for Means - One Group

Large Sample ($n > 30$) – Z-test

Hypotheses:

■ Two-tailed:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$$

■ Right-tailed:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu > \mu_0$$

■ Left-tailed:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu < \mu_0$$

Test Statistic:

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Decision Rule:

- Two-tailed: Reject H_0 if $|Z| > Z_{\alpha/2}$
- Right-tailed: Reject H_0 if $Z > Z_{\alpha}$
- Left-tailed: Reject H_0 if $Z < -Z_{\alpha}$



Test for Means - One Group

Small Sample ($n < 30$) – t-test

Hypotheses:

■ Two-tailed:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$$

■ Right-tailed:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu > \mu_0$$

■ Left-tailed:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu < \mu_0$$

Test Statistic:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad df = n - 1$$

Decision Rule:

- Two-tailed: Reject H_0 if $|t| > t_{\alpha/2, n-1}$
- Right-tailed: Reject H_0 if $t > t_{\alpha, n-1}$
- Left-tailed: Reject H_0 if $t < -t_{\alpha, n-1}$



t-Statistic, Z-Statistic & χ^2 Statistic

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. Define

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Standard normal or Z-Statistic

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Chi-square χ^2 Statistic U

$$U = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$



t-Statistic, Z-Statistic & χ^2 Statistic

t-statistic

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n})}{S/\sigma} = \frac{Z}{S/\sigma}.$$

Since

$$\frac{S}{\sigma} = \sqrt{\frac{(n-1)S^2}{\sigma^2} \cdot \frac{1}{n-1}} = \sqrt{\frac{U}{n-1}},$$

we obtain

$$t = \frac{Z}{\sqrt{U/(n-1)}}$$

where $Z \sim N(0, 1)$ and $U \sim \chi^2_{n-1}$.



t-Statistic, Z-Statistic & χ^2 Statistic

Independence of Z and U

- By **Cochran's theorem**:

$$\underbrace{\sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2}}_{\chi_n^2} = \underbrace{\frac{n(\bar{X} - \mu)^2}{\sigma^2}}_{\chi_1^2} + \underbrace{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2}}_{\chi_{n-1}^2},$$

and these two components are independent.

- The first term corresponds to Z^2 , the second to U .
- Hence Z and U are independent.



t-Statistic, Z-Statistic & χ^2 Statistic

Because $Z \sim N(0, 1)$ and $U \sim \chi_{n-1}^2$ are independent,

$$T = \frac{Z}{\sqrt{U/(n-1)}} \sim t_{n-1}.$$

A Student-t distributed variable t with $\nu = n - 1$ degrees of freedom has a probability density function $f_T(t)$

$$f_T(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad \nu = n - 1.$$

For any complex number z with $\Re(z) > 0$,

$$\Gamma(z) = \int_0^{\infty} x^{z-1} e^{-x} dx$$

$$\Gamma(z+1) = z \Gamma(z)$$



Mean and Variance of Students t Distribution

For a t -distribution with ν degrees of freedom,

$$f(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad -\infty < t < \infty$$

Mean

$$E[t] = \begin{cases} 0, & \nu > 1, \\ \text{undefined}, & \nu \leq 1. \end{cases}$$

Variance

$$\text{Var}(t) = \begin{cases} \frac{\nu}{\nu - 2}, & \nu > 2, \\ \infty, & 1 < \nu \leq 2, \\ \text{undefined}, & \nu \leq 1. \end{cases}$$



Two Group Tests

Independent Sample Tests & Dependent Sample Tests

Test for Means - Two Groups

Assumptions

Applicable to:

- Independent two-sample tests (Z-test or t-test)
- Dependent (paired) sample t-test

General Assumptions:

- **Normality:** Each population from which samples are drawn is normally distributed.
- **Scale:** Data are measured on an interval or ratio scale.
- **Random Sampling:** Samples are randomly and independently drawn from the populations.



Test for Means - Two Groups

Assumptions

For Independent Two-Sample Tests:

- The two samples are independent of each other.
- Population variances are:
 - **Equal** for **pooled t-test** (assume homogeneity of variances).
 - **Unequal** for **unpooled (Welch's) t-test**.
- Sample sizes can be equal or unequal.

For Paired Sample Test:

- Observations are paired (before–after, matched pairs, or repeated measures).
- Differences between pairs are normally distributed.



Independent Samples

Understanding Independent Samples

Independent Samples:

- Two groups are selected independently; observations in one group are not related to those in the other.
- Examples:
 - Comparing average test scores of students from two different schools.
 - Comparing mean income levels of urban and rural households.
 - Measuring the effectiveness of two different fertilizers on separate plots of land.
 - Comparing mean weights of two distinct animal species.



Dependent Samples

Understanding the Dependent Sample

Dependent (Paired) Samples:

- Observations are paired — the same or matched subjects are measured twice or under two conditions.
- Examples:
 - Comparing blood pressure of patients *before* and *after* taking medication - Before/After pairs
 - Measuring students' performance *before* and *after* a training program - Before/After pairs
 - Comparing yields of the same crop under two different irrigation methods on the same plots - Matched Pairs
 - Comparing accuracy of two teaching methods on the same group of students - Matched Pairs



Test for Means - Two Groups

Pooled t -test for Independent Samples (Equal Variances Assumed)

Hypotheses:

■ Two-tailed:

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 \neq \mu_2$$

■ Right-tailed:

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 > \mu_2$$

■ Left-tailed:

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 < \mu_2$$

The difference in sample means, $\bar{X}_1 - \bar{X}_2$, is a random variable with mean $E[\bar{X}_1 - \bar{X}_2] = \mu_1 - \mu_2$ and variance $\text{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$.



Test for Means - Two Groups

Pooled t -test for Independent Samples (Equal Variances Assumed)

Test Statistic:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where

$$S_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \quad df = n_1 + n_2 - 2$$

Decision Rule:

- Two-tailed: Reject H_0 if $|t| > t_{\alpha/2, n_1+n_2-2}$
- Right-tailed: Reject H_0 if $t > t_{\alpha, n_1+n_2-2}$
- Left-tailed: Reject H_0 if $t < -t_{\alpha, n_1+n_2-2}$



Measuring the Effect Size with Cohen's d

Cohen's d : Effect Size Cohen's d measures the **standardized difference** between two means.

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_p}$$

where the pooled standard deviation is

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Interpretation:

- $d = 0.2$: Small effect — difference is minor
- $d = 0.5$: Medium effect — noticeable difference
- $d = 0.8$: Large effect — substantial difference
- $d > 1.2$: Very large effect — very substantial difference



Pooled t -test for Independent Samples (Equal Variances Assumed)

Problem: An education researcher wants to determine whether a new teaching method improves student performance compared to the traditional method. Two independent groups of students were randomly assigned to each teaching method, and their final exam scores were recorded.

New Method: $n_1 = 15$, $\bar{x}_1 = 78.4$, $s_1 = 6.2$

Traditional Method: $n_2 = 14$, $\bar{x}_2 = 72.5$, $s_2 = 5.8$

Assume that population variances are equal. Test at $\alpha = 0.05$ whether the new teaching method leads to a significantly different mean score.



Pooled t -test for Independent Samples (Equal Variances Assumed)

Solution

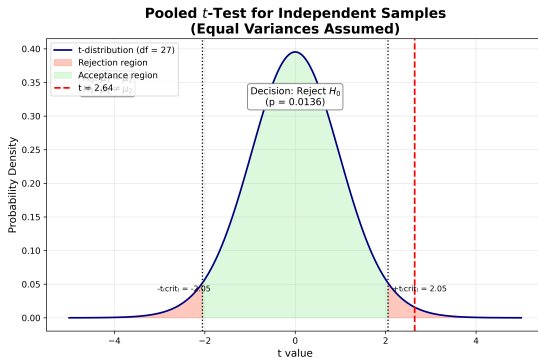


Figure: Pooled t -test



Test for Means - Two Groups

Unpooled t -test for Independent Samples (Unequal Variances)

Hypotheses:

■ Two-tailed:

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 \neq \mu_2$$

■ Right-tailed:

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 > \mu_2$$

■ Left-tailed:

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 < \mu_2$$

The difference in sample means, $\bar{X}_1 - \bar{X}_2$, is a random variable with mean $E[\bar{X}_1 - \bar{X}_2] = \mu_1 - \mu_2$ and variance $\text{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$.



Test for Means - Two Groups

Unpooled t -test for Independent Samples (Unequal Variances)

Test Statistic:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Degrees of Freedom (Welch-Satterthwaite Approximation):

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}$$

Decision Rule:

- Two-tailed: Reject H_0 if $|t| > t_{\alpha/2, df}$
- Right-tailed: Reject H_0 if $t > t_{\alpha, df}$
- Left-tailed: Reject H_0 if $t < -t_{\alpha, df}$



Unpooled t -test for Independent Samples (Unequal Variances)

Problem: A researcher tests whether a new drug reduces symptom severity more effectively than a placebo.

Drug group: $n_1 = 12$, $\bar{x}_1 = 8.5$, $s_1 = 2.1$

Placebo group: $n_2 = 10$, $\bar{x}_2 = 5.2$, $s_2 = 1.8$

Test at $\alpha = 0.05$ whether the drug produces a significantly different mean reduction. Assume that the population variances are very different.



Unpooled t -test for Independent Samples (Unequal Variances)

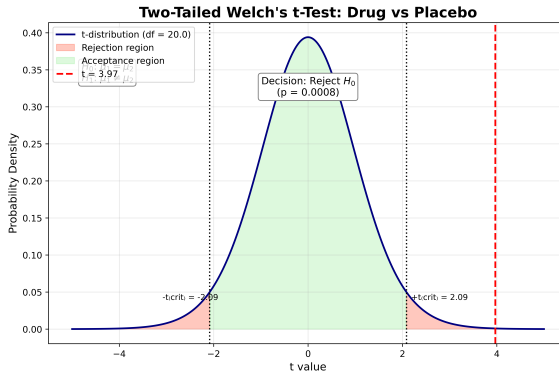


Figure: Welch t-test - Unequal Variance

Frame Title



Test for Means - Two Groups

t-test for Dependent Samples

- **Paired Observations:** Each observation in one group is paired with a specific observation in the other group.
- **Normality of Differences:** The differences $D_i = X_i - Y_i$ are approximately normally distributed.
- **Scale of Measurement:** Data should be interval or ratio scale.
- **Random Sampling:** Pairs are randomly selected from the population.
- **Independence of Pairs:** Each pair is independent of other pairs.

Hypotheses:

- Two-tailed: $H_0 : \mu_D = 0, \quad H_1 : \mu_D \neq 0$
- Right-tailed: $H_0 : \mu_D = 0, \quad H_1 : \mu_D > 0$
- Left-tailed: $H_0 : \mu_D = 0, \quad H_1 : \mu_D < 0$

Difference per pair: $D_i = X_i - Y_i, \quad i = 1, \dots, n$



Test for Means - Two Groups

t-test for Dependent Samples

Sample mean and variance of differences:

$$\bar{D} = \frac{1}{n} \sum_{i=1}^n D_i, \quad s_D^2 = \frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D})^2$$

Test Statistic:

$$t = \frac{\bar{D} - \mu_D}{s_D / \sqrt{n}}, \quad df = n - 1$$

Decision Rule:

- Two-tailed: reject H_0 if $|t| > t_{\alpha/2, n-1}$
- Right-tailed: reject H_0 if $t > t_{\alpha, n-1}$
- Left-tailed: reject H_0 if $t < -t_{\alpha, n-1}$



Paired t -test for Dependent Samples

Problem:

A researcher wants to determine whether an 8-week gym training program significantly improves participants' strength. Ten individuals are measured for their 1-repetition maximum (1RM) bench press **before** and **after** the program. If someone's 1RM on the bench press is 100 kg, it means they can lift 100 kg once

Before training: [100, 105, 98, 110, 102, 95, 108, 101, 99, 104]

After training: [105, 108, 100, 115, 106, 98, 112, 103, 101, 108]

At a significance level of $\alpha = 0.05$, test whether the training program leads to a significant improvement in mean bench press strength.



Paired t -test for Dependent Samples

Solution

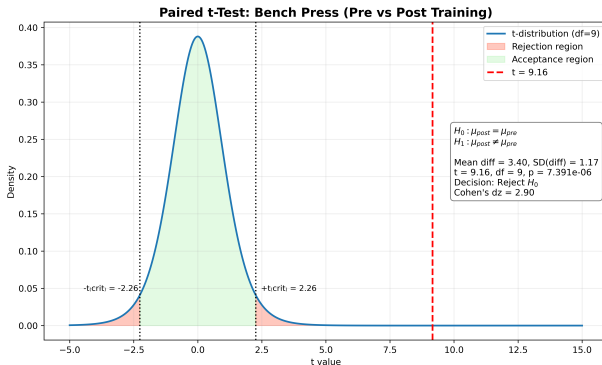


Figure: Paired t -test



Test for Variance

χ^2 & F Tests

One Group Test

χ^2 Test

Chi-square Distribution

Definition

A Chi-square random variable with n degrees of freedom is defined as

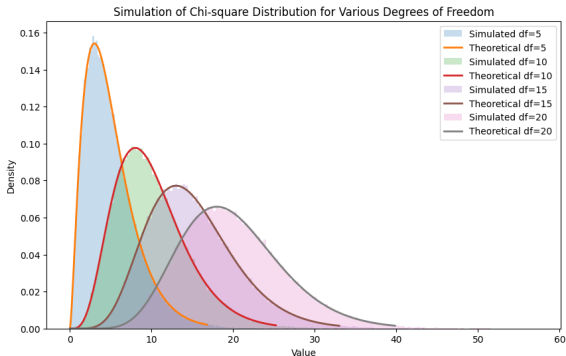
$$\chi_n^2 = \sum_{i=1}^n Z_i^2, \quad Z_i \sim N(0, 1) \text{ i.i.d.}$$

Idea of Simulation

- Generate n independent samples from $N(0, 1)$.
- Square them and sum to obtain one χ_n^2 sample.
- Repeat this many times to approximate the distribution.



Simulation of Chi-square Distribution



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Figure: Simulation of Chi-square Distribution for varying degrees of freedom



Test for Variance - One Group

Chi-Square Test for Variance

- Test for Variance - One Group is a parametric test used to check whether the population variance equals a hypothesized value.
- **Assumptions:**
 - Normality: The population from which the sample is drawn is normally distributed.
 - Scale: Data should be measured at an interval or ratio scale.
 - Random Sampling: Samples are randomly drawn from the population.
 - Independent Observations: Observations are independent of each other.



Test for Variance - One Group

Chi-Square Test for Variance

Hypotheses:

■ Two-tailed:

$$H_0 : \sigma^2 = \sigma_0^2, \quad H_1 : \sigma^2 \neq \sigma_0^2$$

■ Right-tailed:

$$H_0 : \sigma^2 = \sigma_0^2, \quad H_1 : \sigma^2 > \sigma_0^2$$

■ Left-tailed:

$$H_0 : \sigma^2 = \sigma_0^2, \quad H_1 : \sigma^2 < \sigma_0^2$$

Test Statistic:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

s^2 is the sample variance, σ_0^2 is the hypothesized variance and n is the sample size.



Test for Variance - One Group

Chi-Square Test for Variance

Distribution:

$$\chi^2 \sim \chi^2_{(n-1)} \text{ under } H_0$$

Decision Rule:

- Two-tailed: Reject H_0 if $\chi^2 > \chi^2_{\alpha/2, n-1}$ or $\chi^2 < \chi^2_{1-\alpha/2, n-1}$
- Right-tailed: Reject H_0 if $\chi^2 > \chi^2_{\alpha, n-1}$
- Left-tailed: Reject H_0 if $\chi^2 < \chi^2_{1-\alpha, n-1}$

α is the significance level and n is the sample size.



Test for Variance - One Group

Chi-Square Test for Variance

Problem:

A biscuit manufacturer claims that the standard deviation of the weight of biscuits in a pack is 2 grams. To verify this claim, a quality control engineer randomly selects 15 biscuits and measures their weights (in grams):

50, 49, 52, 51, 50, 48, 53, 49, 51, 50, 52, 48, 49, 51, 50

Test, at $\alpha = 0.05$, whether the **variance of biscuit weights** differs from the claimed variance of $\sigma^2 = 4 \text{ g}^2$.



Test for Variance - One Group

Solution

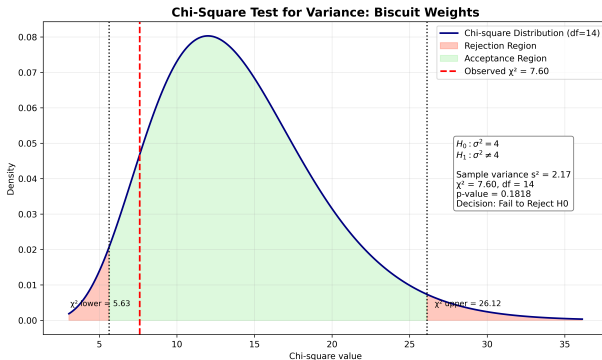


Figure: Chi-Square Test for Variance



Two Group Test

F Test

Test for Variances - Two Groups

F-test for Comparing Two Population Variances

- The F-test is used to test whether the variances of two populations are equal.
- **Assumptions:**
 - Normality: Both populations are normally distributed.
 - Scale: Data should be measured at interval or ratio scale.
 - Random Sampling: Samples are randomly and independently drawn.
 - Independent Observations: Samples from the two groups are independent of each other.



Test for Variances - Two Groups

F-test for Comparing Two Population Variances

Hypotheses:

■ Two-tailed:

$$H_0 : \sigma_1^2 = \sigma_2^2, \quad H_1 : \sigma_1^2 \neq \sigma_2^2$$

■ Right-tailed:

$$H_0 : \sigma_1^2 = \sigma_2^2, \quad H_1 : \sigma_1^2 > \sigma_2^2$$

■ Left-tailed:

$$H_0 : \sigma_1^2 = \sigma_2^2, \quad H_1 : \sigma_1^2 < \sigma_2^2$$

Test Statistic:

$$F = \frac{s_1^2}{s_2^2}$$

s_1^2 is larger sample variance and s_2^2 is smaller sample variance



Test for Variances - Two Groups

F-Statistic and χ^2 Statistic

Let

$$X_1, X_2, \dots, X_{n_1} \sim N(\mu_1, \sigma_1^2), \quad Y_1, Y_2, \dots, Y_{n_2} \sim N(\mu_2, \sigma_2^2)$$

be two independent samples. We want to test:

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad \text{vs.} \quad H_1 : \sigma_1^2 \neq \sigma_2^2$$

$$s_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2, \quad s_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$
$$\frac{(n_1 - 1)s_1^2}{\sigma_1^2} \sim \chi_{n_1-1}^2, \quad \frac{(n_2 - 1)s_2^2}{\sigma_2^2} \sim \chi_{n_2-1}^2$$



Test for Variances - Two Groups

F-Statistic and χ^2 Statistic

F-Statistic:

$$F = \frac{\frac{(n_1-1)s_1^2}{\sigma_1^2} / (n_1 - 1)}{\frac{(n_2-1)s_2^2}{\sigma_2^2} / (n_2 - 1)} = \frac{s_1^2 / \sigma_1^2}{s_2^2 / \sigma_2^2}$$

If $H_0 : \sigma_1^2 = \sigma_2^2 = \sigma^2$, then

$$F = \frac{s_1^2}{s_2^2} \sim F_{n_1-1, n_2-1}$$

Note: $n_1 - 1$ and $n_2 - 1$ are the numerator and denominator degrees of freedom, respectively.



Test for Variances - Two Groups

F-test for Comparing Two Population Variances

- Under $H_0 : \sigma_1^2 = \sigma_2^2$, the test statistic follows an F-distribution with $(n_1 - 1, n_2 - 1)$ degrees of freedom:

$$F \sim F_{n_1-1, n_2-1}$$

■ Decision Rule:

- Two-tailed: Reject H_0 if $F < F_{1-\alpha/2, n_1-1, n_2-1}$ or $F > F_{\alpha/2, n_1-1, n_2-1}$
- Right-tailed: Reject H_0 if $F > F_{\alpha, n_1-1, n_2-1}$
- Left-tailed: Reject H_0 if $F < F_{1-\alpha, n_1-1, n_2-1}$



F-Test for Equality of Variances

Problem:

A biscuit company produces biscuits in two different factories, Factory A and Factory B. The quality control team wants to check if the variability in biscuit weights is the same for both factories.

A random sample of biscuit weights (in grams) from each factory is collected:

Factory A ($n_1 = 10$): 50, 52, 49, 51, 50, 48, 53, 49, 51, 50

Factory B ($n_2 = 12$): 47, 50, 49, 48, 51, 46, 52, 49, 48, 50, 47, 49

Test, at $\alpha = 0.05$, whether the **variance of biscuit weights** is the same in the two factories.



F-Test for Equality of Variances

Solution

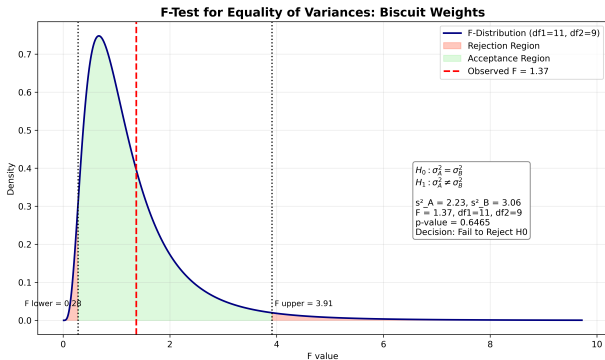


Figure: F-test



Test for Proportion

χ^2 & F Tests

One Group Test

Z, Chi-Square for Independence Test

Test for Proportion - One Group

Z-test

- Test for Proportion - One Group determine whether the proportion of successes in a population differs significantly from a hypothesized value.
- Assumptions
 - **Random Sampling:** The sample must be randomly selected from the population.
 - **Independent Observations:** Each observation should be independent of the others.
 - **Sample Size Adequacy:** The sample size should be large enough for the normal approximation to be valid:

$$np_0 \geq 5 \quad \text{and} \quad n(1 - p_0) \geq 5$$

where p_0 is the hypothesized population proportion.

- **Binary Outcome:** The variable under study should be categorical with two possible outcomes (e.g., success/failure, pass/fail).



Hypotheses for One-Sample Proportion Test

Z-test

- **Two-tailed:**

$$H_0 : p = p_0, \quad H_1 : p \neq p_0$$

- **Right-tailed:**

$$H_0 : p = p_0, \quad H_1 : p > p_0$$

- **Left-tailed:**

$$H_0 : p = p_0, \quad H_1 : p < p_0$$



Hypotheses for One-Sample Proportion Test

Z-test

Test Statistic:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where $\hat{p} = \frac{x}{n}$ is the sample proportion and n is the sample size.

Decision Rule:

- Two-tailed: Reject H_0 if $|Z| > Z_{\alpha/2}$
- Right-tailed: Reject H_0 if $Z > Z_{\alpha}$
- Left-tailed: Reject H_0 if $Z < -Z_{\alpha}$



Hypotheses for One-Sample Proportion Test

Z-test

Let $X_1, X_2, \dots, X_n$ be i.i.d. Bernoulli random variables, where

$$X_i = \begin{cases} 1 & \text{if "success"} \\ 0 & \text{if "failure"} \end{cases}$$

The probability of success is p_0 .

Sample proportion:

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$$

$\hat{p}$ is approximately normally distributed with

$$E[\hat{p}] = p_0, \quad \text{Var}(\hat{p}) = \frac{p_0(1 - p_0)}{n}, \quad \text{for large } n.$$



Hypotheses for One-Sample Proportion Test

Z-test

Step 1: Mean and variance of $\hat{p}$:

$$E[\hat{p}] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \cdot np_0 = p_0$$

$$Var(\hat{p}) = \frac{1}{n^2} \sum_{i=1}^n Var(X_i) = \frac{1}{n^2} \cdot np_0(1 - p_0) = \frac{p_0(1 - p_0)}{n}$$



Hypotheses for One-Sample Proportion Test

Z-test

Step 2: Apply Central Limit Theorem (CLT):

For large n , the sum of i.i.d. variables $\sum X_i$ is approximately normal:

$$\sum_{i=1}^n X_i \sim N(np_0, np_0(1-p_0))$$

Dividing by n :

$$\hat{p} = \frac{1}{n} \sum X_i \sim N\left(p_0, \frac{p_0(1-p_0)}{n}\right)$$

We standardize $\hat{p}$ to create a Z-statistic:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$Z \sim \mathcal{N}(0, 1)$$



Decision Making in One-Sample Proportion Test

Using Z-Statistics

Step 3: Define significance level and critical region

- Choose a significance level α (commonly 0.05).
- Determine the type of test:
 - **Two-tailed:** $H_0 : p = p_0$ vs $H_1 : p \neq p_0$ Reject H_0 if $|Z| > Z_{\alpha/2}$
 - **Left-tailed:** $H_0 : p = p_0$ vs $H_1 : p < p_0$ Reject H_0 if $Z < -Z_{\alpha}$
 - **Right-tailed:** $H_0 : p = p_0$ vs $H_1 : p > p_0$ Reject H_0 if $Z > Z_{\alpha}$

Step 4: Compare calculated Z with critical Z

If Z_{calc} falls in the rejection region, reject H_0

Step 5: Compute p-value (optional)

- $p\text{-value} = P(|Z| \geq |Z_{\text{calc}}|)$ for two-tailed
- Reject H_0 if $p\text{-value} < \alpha$



Hypotheses for One-Sample Proportion Test

Z-test

A university claims that only 10% of its students use illicit drugs. To verify this claim, a survey is conducted on 200 randomly selected students. It is found that 28 students reported using drugs. Test, at a significance level of $\alpha = 0.05$, whether the true proportion of students using drugs is different from 0.10.



Hypotheses for One-Sample Proportion Test

Z-test

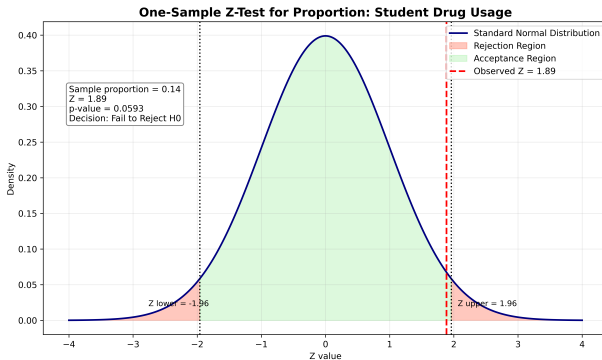


Figure: Z-test for One Group Proportion



Two Group Test

Z-Test for Proportions

Test for Proportion - Two Groups

Z-test for Difference in Proportions

- The **two-sample proportion Z-test** determines whether the proportions of successes in two independent populations differ significantly.

- **Assumptions:**

- **Independent Samples:** The two samples must be drawn independently.
- **Random Sampling:** Each sample should be randomly selected.
- **Sample Size Adequacy:** Normal approximation is valid if

$$n_1\hat{p}_1 \geq 5, \quad n_1(1 - \hat{p}_1) \geq 5, \quad n_2\hat{p}_2 \geq 5, \quad n_2(1 - \hat{p}_2) \geq 5$$

- **Binary Outcome:** Each observation represents a success or failure.



Hypotheses for Two-Sample Proportion Test

■ Two-tailed Test:

$$H_0 : p_1 = p_2 \quad \text{vs.} \quad H_1 : p_1 \neq p_2$$

■ Left-tailed Test:

$$H_0 : p_1 = p_2 \quad \text{vs.} \quad H_1 : p_1 < p_2$$

■ Right-tailed Test:

$$H_0 : p_1 = p_2 \quad \text{vs.} \quad H_1 : p_1 > p_2$$



Test Statistic and Decision Rule for Two-Sample Proportion Test

■ Test Statistic:

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where

$$\hat{p}_i = \frac{x_i}{n_i}, \quad \hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

■ Decision Rule:

- **Two-tailed test:** Reject H_0 if $|z| > z_{\alpha/2}$
- **Right-tailed test:** Reject H_0 if $z > z_{\alpha}$
- **Left-tailed test:** Reject H_0 if $z < -z_{\alpha}$



Multiple Group Test

Chi-Square Test for Independence or Homogeneity

Test for Proportion - Multiple Groups

Chi-Square Test for Independence or Homogeneity

- Test for Proportion - Multiple Groups checks whether two or more categorical variables are statistically independent or whether the distribution of proportions across categories is the same for all groups.
- **Assumptions**
 - **Random Sampling:** The data should be obtained from a random sample of the population.
 - **Independence of Observations:** Each observation must belong to one and only one cell of the contingency table. No individual or item should be counted more than once.
 - **Expected Frequency Condition:** The expected frequency in each cell should be sufficiently large for the chi-square approximation to the sampling distribution to be valid:

$$E_{ij} \geq 5 \quad \text{for most cells.}$$

- **Measurement Scale:** Both variables should be categorical (nominal or ordinal).



Hypotheses for Proportion - Multiple Groups

Chi-Square Test for Independence or Homogeneity

■ Hypotheses:

H_0 : The two categorical variables are independent

H_1 : The two categorical variables are not independent

■ Hypotheses:

$H_0 : p_1 = p_2 = p_3 = \cdots = p_k$

H_1 : At least one proportion p_i differs.



Hypotheses for Proportion - Multiple Groups

Chi-Square Test for Independence or Homogeneity

■ Test Statistic:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where

$$E_{ij} = \frac{(R_i)(C_j)}{N}$$

O_{ij} : observed frequency,

E_{ij} : expected frequency,

R_i : total of row i ,

C_j : total of column j ,

N : grand total

r : number of rows

c : number of columns



Hypotheses for Proportion - Multiple Groups

Chi-Square Test for Independence or Homogeneity

■ Observed Frequency Matrix:

$$O = \begin{bmatrix} O_{11} & O_{12} & \dots & O_{1c} \\ O_{21} & O_{22} & \dots & O_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ O_{r1} & O_{r2} & \dots & O_{rc} \end{bmatrix}$$

■ Expected Frequency Matrix:

$$E = \begin{bmatrix} E_{11} & E_{12} & \dots & E_{1c} \\ E_{21} & E_{22} & \dots & E_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ E_{r1} & E_{r2} & \dots & E_{rc} \end{bmatrix}$$



Hypotheses for Proportion - Multiple Groups

Chi-Square Test for Independence or Homogeneity

- The chi-square test with degrees of freedom $df = (r-1)(c-1)$ compares **observed frequencies** O_{ij} with **expected frequencies** E_{ij} under the null hypothesis H_0 of independence:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

- Under H_0 , the counts O_{ij} in each cell can be approximated by a **multinomial distribution**.
- Each standardized term

$$\frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}$$

is approximately Normal(0, 1) for large sample sizes ($E_{ij} \geq 5$).

- The sum of the squares of $r \times c$ independent standard normal variables is **Chi-square distributed**.



Hypotheses for Proportion - Multiple Groups

Chi-Square Test for Independence or Homogeneity

- Consider a contingency table with r rows and c columns. Each observation falls into exactly one cell (i, j) .
- Let N be the total number of observations and p_{ij} the probability of an observation being in cell (i, j) . Then the vector of cell counts:

$$(O_{11}, O_{12}, \dots, O_{rc}) \sim \text{Multinomial}(N, (p_{11}, p_{12}, \dots, p_{rc}))$$

- Under the null hypothesis of independence:

$$H_0 : p_{ij} = \frac{R_i C_j}{N^2} \quad \Rightarrow \quad E_{ij} = N p_{ij} = \frac{R_i C_j}{N}$$

where R_i and C_j are the row and column totals, respectively.

- The Chi-square test statistic χ^2 compares observed counts O_{ij} to expected counts E_{ij} under the multinomial model.



Hypotheses for Proportion - Multiple Groups

Chi-Square Test for Independence or Homogeneity

- Under the null hypothesis H_0 of independence, the chi-square statistic is

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

For large sample size N , this statistic approximately follows a chi-square distribution with $(r - 1)(c - 1)$ degrees of freedom:

$$\chi^2 \sim \chi_{(r-1)(c-1)}^2.$$

The degrees of freedom arise because:

$$\text{df} = rc - 1 - [(r - 1) + (c - 1)] = (r - 1)(c - 1).$$

- **Decision Rule:** Reject H_0 if

$$\chi_{\text{calc}}^2 > \chi_{\alpha, (r-1)(c-1)}^2$$



Hypotheses for Proportion - Multiple Groups

Chi-Square Test for Independence or Homogeneity

- **Question:** A random sample of 500 U.S. adults was surveyed regarding their political affiliation and opinion on a tax reform bill. Does one's opinion depend on party affiliation?
- **Observed Data (Contingency Table):**

$O =$	Party / Opinion	Favor	Indifferent	Opposed	Total
	Democrat	138	83	64	285
	Republican	64	67	84	215
	Total	202	150	148	500

- **Null Hypothesis (H_0):**

H_0 : Opinion on the tax reform bill is independent of political affiliation.

- **Alternative Hypothesis (H_1):**

H_1 : Opinion on the tax reform bill depends on political affiliation.



Hypotheses for Proportion - Multiple Groups

Chi-Square Test for Independence or Homogeneity

Observed counts:

$O =$	Party / Opinion	Favor	Indifferent	Opposed	Row Total
	Democrat	138	83	64	285
	Republican	64	67	84	215
	Column Total	202	150	148	500

Expected counts under H_0 :

$E =$	Party / Opinion	Favor	Indifferent	Opposed	Row Total
	Democrat	115.14	85.50	84.36	285
	Republican	86.86	64.50	63.64	215
	Column Total	202	150	148	500

$$\chi^2 = \sum_{i=1}^2 \sum_{j=1}^3 \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = 22.152, \quad df = 2$$

Critical value: $\chi_{0.05,2}^2 = 5.991$

Since $22.152 > 5.991$, we **reject** H_0 .



References I



Douglas C Montgomery and George C Runger.
Applied statistics and probability for engineers.
John wiley & sons, 2010.

